GE Hydro

INSPECTION AND TEST PLAN

Section 1900

GOVERNOR SYSTEM

Project: SEVEN MILE

Unit 4

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1901

CODE A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
1901	100	I,R	Incoming inspection.	See ITP Section 100		QCR 101	
1902	2300	O	Install accumulator air/oil tank. Install the governor oil sump tank.	Installation proc. SMI-U04-03024 Steps 1.1 and 1.2 Drawing 222038288			Jan 28th 2003
1903	2300	I,R	Perform a visual inspection on all parts of the piping installation.	Installation proc. SMI-U04-03024 Step 1.3 Drawing 222038288		QCR 004√	Feb 12th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1902

CODE
A: Hold point/ customer
T: Witness point/ customer

I : Inspection and test point
R: Results recorded in Q-A files
Copy to customer

M: Monitoring in process
O: Assembly Operation
V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
1904	2300	O	Proceed to the trial installation of the governor system oil piping. Prepare and proceed to the acid pickling of the servomotor piping.	Installation proc. SMI-U04-03024 Steps 1.4 to 1.6 Drawing 222038288			Feb 1st 2003
1905	2300	O	Install wiring. Fill the sump tank with clean filtred governor oil. Perform the pump rotation check.	Installation proc. SMI-U04-03024 Steps 1.7 to 1.9 Drawing 222038288			Feb 12th 2003.
1906	2300	O	After pickling, prepare the piping for flushing operation. Perform the flushing and reconnect the inlet and outlet pipes to the servomotor ports.	Installation proc. SMI-U04-03024 Steps 1.10 and 1.11			Feb 7th 03

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1903

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Drawing 222038288			
1907	2300	O,M,R	Perform a pressure test on the section of piping between item 119 and 200.	Installation proc. SMI-U04-03024 Step 1.12 Drawing 222038288		QCR 005 ✓ See step 1915	Feb 19th 2003
1908	2300	O	Check the outside surfaces of the sump tank and clean thoroughly.	Installation proc. SMI-U04-03024 Step 1.13 Drawing 222038288			Feb 9th 2003.
1909	2300	I,R	Verify the cleanliness of the inside of the sump tank.	Installation proc. SMI-U04-03024 Step 1.13		QCR 004	Feb 9th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1904

CODE
A: Hold point/ customer
T: Witness point/ customer

I : Inspection and test point
R: Results recorded in Q-A files
Copy to customer

M: Monitoring in process
O: Assembly Operation
V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Drawing 222038288			/
1910	2300	O	Check the outside surfaces of the accumulator tank and clean thoroughly.	Installation proc. SMI-U04-03024 Step 1.14 Drawing 222038288			Feb 9th 2003
1911	2300	I,R	Verify the cleanliness of the inside of the accumulator tank.	Installation proc. SMI-U04-03024 Step 1.14 Drawing 222038288		QCR 004 ✓	Feb 8th 2003
1912	2300	O,M,R	Install a new manhole gasket on the accumulator tank, then seal the manhole opening. Torque.	Installation proc. SMI-U04-03024 Step 1.15	Torque value is 1965 N.m lub.	QCR 004 ✓	Feb 13th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1905

CODE		
A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
			Torque monitoring	Drawing SD0222615400-04			
1913	2300	O	Set the servomotor cushioning valve in halfway open position. Ensure that the drain valves are closed and the servomotor stroke locking device is disengaged. Transfer oil from the sump tank into accumulator tank and raise the level to the high oil level alarm.	Installation proc. SMI-U04-03024 Steps 1.16 to 1.18 Drawing 222038288			Feb 12th 2003.
1914	2300	O	Pressurize the governor system and check for leaks during the pressurizing. Correct detected leaks immediately.	Installation proc. SMI-U04-03024 Steps 1.19 to 1.24 Drawing 222038288			Feb 12th 2003
1915	2300	O,M,R	Perform a pressure test on the governor system piping.	Installation proc. SMI-U04-03024 Section 2.0		QCR 005 ✓	Feb 12th 2003.

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro	INSPECTION AND TEST PLAN	
Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1900 : Governor system	Page: 1906

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Drawing 222038288			
1916	2300	I,R	Final inspection : Ensure that all the work and documentation have been completed.	Installation proc. SMI-U04-03024		QCR 003	04/28/03

RECEIVED 28 Apr. 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 13th, 2003	Accepted by: Date:	Revision no: 00

	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1903

EQUIPMENT : **Governor system piping visual inspection**

A visual inspection of the governor system piping has been done. The pipes were clean.

RECEIVED FEB 12 2003 CA

Measured by.: F. Boulet (Feb 5th 1/2003)

GE Repres.: Philippe Chamontagne (Feb 10th 2003)

Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: ______ Comments: ______	Rev.: 0 Date: 07/23/01

GE	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1909

EQUIPMENT : **Governor system sump tank cleanliness.**

A visual inspection of the sump tank has been done. The tank was clean and no contaminants have been seen.

Measured by.:	F. Boulet	(Feb 5th, 2003)
GE Repres.:	[signature]	(Feb 7th, 2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 07/23/01

GENERAL INSPECTION REPORT	QCR 004
PROJECT: Seven Mile Unit: 4	ITP step no.: 1911

EQUIPMENT : **Governor system accumulator tank cleanliness**

A visual inspection of the accumulator tank has been done. The tank was clean and no foreign contaminants have been observed.

Measured by.:	F. Boulet	(Feb/5th/2003)
GE Repres.:		(Feb/8th/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: ____ Comments: ____	Rev.: 0 Date: 07/23/01

GENERAL INSPECTION REPORT	QCR 004
PROJECT: Seven Mile Unit: 4	ITP step no.: 1912

EQUIPMENT : **Torquing of accumulator tank door**

The accumulator tank manhole opening door has been torqued at a value of 1965 N.m lubricated.
The gasket has been changed.

RECEIVED FEB 1 2 2003

Measured by.: F. Boulet (Feb 6th/2003)
GE Repres.: (Feb 9th/2003)
Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 07/23/01

	PRESSURE TEST	QCR 005
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1915 and 1907

VERIFIED EQUIPMENT(S): Governor system piping (main lines to the servomotors and gate lock lines)

TEST DESCRIPTION : The governor piping has been pressurized with the jockey pump. This has been done in the closed and opened position. The test has been performed with oil.

PRESSURE : 10,3 Mpa (1500 PSI) **DURATION :** 1 hour in closed and opened position

ACCEPTED TEST	X
REJECTED TEST	

COMMENTS :

RECEIVED FEB 12 2003

Measured by.: (Feb/ 9th/2003)

GE Repres.: (Feb/ 9th/2003)

Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 23/07/02

	PRESSURE TEST	QCR 005
	PROJECT: Seven Mile Unit: 4	ITP step no.: N/A

ITP 1900.

VERIFIED EQUIPMENT(S): HPU air lines
All the air line from the two compressors to the receiver and the accumulator, excluding compressors, accumulator and receiver.

TEST DESCRIPTION : 1 hour at 2250 psi with argon.

PRESSURE : 2250 psi **DURATION :** 1 hour

ACCEPTED TEST [X]
REJECTED TEST [ ]

COMMENTS :

RECEIVED April 28/03

Measured by.: Cliff Treman (04/23/03)
GE Repres.: [signature] (04/23/03)
Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 23/07/02

	FINAL INSPECTION	QCR 003
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1916

THIS FORM MUST BE COMPLETED BY THE TECHNICAL DIRECTOR, QUALITY ASSURANCE OR HIS REPRESENTATIVE.

THIS FORM MUST BE COMPLETED AT THE END OF WORK FOR EACH SECTION IN THE INSPECTION AND THE TEST PLAN.

ALL NON CONFORMANCES, DISCREPANCIES, ETC. MUST BE CORRECTED BEFORE COMPLETING THIS FORM.

ONLY THE SPECIFIED ITEMS IN CONTRACT ARE SUBMITTED TO THE FINAL INSPECTION.

	YES	NO	N/A
ALL WORK WAS COMPLETED IN CONFORMITY WITH INSTALLATION MANUAL AND DRAWINGS.	✓		
THE FINAL PRODUCT HAVE BEEN INSPECTED AT THE END OF WORK AND IS IN CONFORMITY WITH CONTRACT REQUIREMENTS.	✓		
ALL NON CONFORMANCES HAVE BEEN CORRECTED ACCORDING TO THE DISPOSITION INSTRUCTIONS.	✓		
ALL CORRECTIVE ACTION REQUESTS HAVE BEEN CORRECTED AT THE CUSTOMER SATISFACTION.	✓		
ALL QUALITY ASSURANCE REPORTS HAVE BEEN COMPLETED ACCORDING TO I.T.P.	✓		
ALL FORMS HAVE BEEN REVISED AND APPROVED BY THE QUALITY ASSURANCE REPRESENTATIVE.	✓		
APPROVAL OF ENGINEERING HAVE BEEN OBTAINED WHEN NECESSARY.	✓		
APPROVAL OF CUSTOMER OR HIS REPRESENTATIVE HAVE BEEN OBTAINED WHEN NECESSARY.	✓		

REF. N.C. AND C.A.R. NB.: NC-SMI-U04-028, NC-SMI-U04-032
NC-SMI-U04-036

COMMENTS :

RECEIVED
5 Apr 2003
CW

Measured by.: [signature] (03/30/03)

GE Repres.: [signature] (03/30/03)

Cust. Repres.: [signature] (/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 07/23/01